

G1 approval field dictionary and validation rules - 2026-08-19

Identity, authority, evidence and decision completion controls

G1 APPROVAL FINALIZATION

Prepared 2026-08-19

Evidence-bound | approval-gated | fail-closed

No production claim without authentic evidence.

Purpose

This is the field-level acceptance contract for every one of the 33 G1 decision-role obligations. It explains what each value means, where it must come from, who owns it, how it is validated and how to correct a failure.

The machine-readable preparation rows are in

G1_33_OBLIGATION_FINALIZATION_REGISTER_2026-08-19.json. No field marked human, approval-event or verifier input may be guessed or generated merely to satisfy the validator.

Field modes

Automatic	Deterministically copied or computed from authoritative bytes/system state
Authoritative retrieval	Read from an organization-approved identity/authority source, then verified
Governance confirmation	A qualified human owner must make or confirm the appointment/policy
Deliberate human action	A real authorized person must approve or reject after fresh authentication
Independent verification	A separate verifier must reproduce and confirm the evidence

Contract-binding fields

decisionId	String; yes	Frozen contract; gate owner	Exactly one of D-01 through D-11; appears once at decision level	Unknown/duplicate ID: discard record and recreate from frozen contract	Automatic
decisionTitle	String; yes for packet	Frozen contract; gate owner	Exact title from matching decision	Mismatch: recopy from contract; do not hand-edit	Automatic
selectedOption	String; yes	Frozen contract; decision owner	Exact byte-for-byte option for the decision	Any mismatch receives no gate credit; recopy from unchanged contract	Automatic
contractArtifactId	String; yes	Frozen contract; gate owner	Equals STOQUIFY-POS-G1-CONTRACT-FREEZE-0.2.0-20260817	Mismatch: stop; bind to correct artifact	Automatic
contractVersion	String; yes	Frozen contract; gate owner	Equals 0.2.0 for this freeze	Mismatch: stop and investigate artifact version	Automatic
contractPath	Repository-relative path; yes	Frozen contract/register; gate owner	Exact normalized path; must resolve within repository	Missing/outside path: reject; use canonical repository-relative path	Automatic
contractSha256	Lowercase 64-hex string; yes	Exact frozen bytes; gate owner/verifier	Recompute SHA-256; must equal 11434eb...a36db	Drift: invalidate all approvals and create a new governed freeze	Automatic plus independent verification

Approver identity fields

<code>accountableApprover</code>	Non-empty display name; yes at approval	Canonical identity/authority roster; governance owner	Must describe the same person as <code>stableSubjectId</code> ; never the identity key	Typed/ambiguous name: reject and resolve through authoritative directory	Authoritative retrieval
<code>stableSubjectId</code>	Immutable string; yes for assurance even though current script does not inspect it	Organization IdP/HR identity master; security/HR owner	Unique, stable, non-display identifier; resolves to signer; tenant appropriate	Missing/mutable ID: reject; provide IdP subject or approved identity ID	Authoritative retrieval
<code>identityProvider</code>	Controlled provider code/URI; yes	Security-approved identity configuration	Provider must be organization-controlled and resolvable	Unknown provider: reject; approve provider or use accepted IdP	Governance confirmation plus retrieval
<code>organizationId</code>	Stable organization ID; yes	Tenant/authority source; governance owner	Must be the organization granting authority	Wrong/missing organization: reject and reissue appointment	Authoritative retrieval
<code>tenantId</code>	Stable tenant ID; yes	Auth session and policy; security owner	Must match G1 approval tenant and authentication organization	Cross-tenant mismatch: reject; never translate IDs manually	Automatic/retrieval
<code>membershipId</code>	Stable membership reference; conditional	Organization membership system; security owner	Active membership for signer/tenant at approval time	Missing when policy requires it: resolve active membership	Authoritative retrieval

Role and authority fields

approverRole	Exact string; yes	Frozen contract and authority roster	Case-insensitive match to one required role; one entry per exact role	Alias/ordinary app role: reject; use exact G1 authority label	Automatic plus governance confirmation
authorityReference	Durable reference; yes	Canonical authority/delegation system; governance owner	Resolves to signer, exact role, tenant, scope and active dates	Locally invented/unresolvable value: reject; issue real appointment reference	Governance confirmation/retrieval
authoritySourcePathOrUri	Path/URI; yes	Approved roster source; governance owner	Resolves to authoritative record; local files also require content hash	Broken/mutable-only source: reject; use controlled source and hash/export	Authoritative retrieval
authorityAssignmentId	Stable ID; yes when registry supports it	Authority registry; governance owner	Unique, tenant-scoped, non-reused appointment identifier	Missing/reused ID: reissue controlled assignment	Authoritative retrieval
delegationReference	Durable reference; conditional	Delegation register; delegating authority	Required when authority is delegated; must specify delegator, scope and expiry	Informal delegation: reject; record bounded formal delegation	Governance confirmation
authorityEffectiveFrom	ISO-8601 timestamp; yes	Authority record; governance owner	At/before approval time	Future start: signer not authorized; wait or correct source error	Authoritative retrieval
authorityEffectiveTo	ISO-8601 timestamp or governed no-expiry value; yes by policy	Authority record; governance owner	After approval and review period, unless policy permits earlier expiry with invalidation	Expired/ambiguous: renew or appoint another person	Authoritative retrieval
authorityVerificationAt	ISO-8601 timestamp; yes	Authority verifier clock	At/after source retrieval and before approval import	Missing/stale verification: reverify	Independent verification
authorityVerificationBy	Stable verifier identity; yes	Verification record; governance owner	Not an anonymous agent/display-name-only value	Missing/unproven verifier: appoint and record qualified verifier	Independent verification
segregationOfDutiesResult	Enum PASS, FAIL, REVIEW_REQUIRED; yes	Approved SoD policy; governance/control owner	Must be PASS for import; evaluates same-person and producer/verifier combinations	Fail/unknown: reassign roles or obtain approved exception where policy allows	Governance confirmation plus independent verification
conflictOfInterestResult	Enum PASS, FAIL, REVIEW_REQUIRED; yes	Conflict policy/declaration; governance owner	Must be PASS or approved documented exception	Missing/fail: resolve conflict or reassign signer	Governance confirmation

Decision fields

approvalStatus	Enum; yes	Controlled workflow; signer/verifier	Working packet stays pending; live decision equals APPROVED only after all three roles verify	Premature APPROVED: revert to pending and complete evidence	Deliberate human action plus verifier
decisionIntent	Enum APPROVE or REJECT; yes	Signer action	Must be deliberate, recorded after fresh auth and bound to decision/hash	Implicit intent/page view: reject; obtain explicit action	Deliberate human action
rationale	Non-empty string; yes	Accountable decision record; decision owners	Must explain why option is accepted with limitations/evidence	Draft/copied generic text only: accountable owners confirm or replace	Deliberate human action
effectiveVersion	Non-empty version string; yes	Contract/deployment policy; decision owner	Valid version in which decision applies; current preparation value 0.2.0	Missing/incorrect: governance confirms exact effective version	Automatic plus governance confirmation
reviewOrExpiryAt	ISO-8601 timestamp; yes	Governance policy; decision owner	Parseable, future at approval time and policy-compliant	Null/expired: set approved review date or reapprove	Governance confirmation
evidenceLinks	Non-empty array; yes	Contract/template plus signed evidence; evidence owner	Every reference resolves; local paths are content-hash checked	Broken/stale link: replace with durable evidence and reverify	Automatic collection plus human review
affectedCapabilities	Non-empty unique string array; yes	Contract/template; product/domain owners	Complete for decision; values normalized and not empty	Missing capability: correct packet and reapprove if signed scope changes	Automatic preparation plus human confirmation
rollbackOrDisablePolicy	Non-empty string; yes	Contract/template; domain/control owners	Fail-closed, operationally actionable and accepted by all decision owners	Vague/missing: rewrite before signing	Governance confirmation

Fresh-authentication fields

authenticationMethod	Controlled string; yes	Trusted identity workflow	Organization-approved method; repository currently supports tenant-bound password assurance	Unknown/self-declared method: reject; use approved workflow	System generated
authenticationAssuranceLevel	Number/controlled enum; yes	Trusted auth service; security owner	Meets approved minimum; current <code>requireFreshAuth</code> minimum is password level	Below level: require stronger reauthentication	System generated
mfaMethod	Controlled string; conditional/preferred	Trusted auth provider	Required if governance policy mandates MFA; must match provider audit	Typed MFA claim: reject; obtain provider event	System generated
authenticationOrganizationId	Stable tenant ID; yes	Auth assurance record	Equals signer authority tenant and G1 tenant	Wrong tenant/null: reject and reauthenticate in correct tenant	System generated
authenticationSessionId	Redacted durable reference; yes for audit, never raw token	Auth service	Resolves internally without exposing session token; session active/not revoked	Raw token exposure: revoke/redact; missing ref: reauthenticate	System generated
freshAuthenticatedAt	ISO-8601 timestamp; yes	Trusted workflow clock	Valid, not future-dated, tenant-bound and recent	Missing/stale/future: reauthenticate	System generated
authenticationEvidenceReference	Durable reference; yes	Auth/audit service	Resolves to method, assurance, subject, tenant and timestamp	Typed/local value: reject; export controlled audit reference	System generated

The repository function `requireFreshAuth(300)` rejects missing, stale, future-dated, wrong-tenant and below-level assurance. The G1 gate separately permits at most ten minutes between `freshAuthenticatedAt` and `approvedAt`. Governance should choose the stricter applicable rule; five-minute fresh assurance plus approval within ten minutes is acceptable only if the workflow keeps the original assurance event trustworthy.

Signature and evidence fields

approvedAt	ISO-8601 timestamp; yes	Trusted workflow clock	At/after fresh auth and no more than ten minutes later	Outside window: reauthenticate and approve again	System generated by deliberate action
signatureReference	Durable provider-neutral reference; yes	Approval/e-sign workflow	Resolves to immutable approval record and audit trail	Invented/broken URI: reject; export real envelope reference	System generated
signatureProvider	Controlled provider code; yes	Approved workflow configuration	Provider is organization-approved and verification method documented	Unknown provider: reject or approve provider through governance	Governance plus system
signatureEnvelopeId	Stable envelope/record ID; yes	Signing/approval provider	Unique, immutable and corresponds to signer/decision/hash	Missing/reused ID: export correct final envelope	System generated
signatureEvidencePathOrUri	Path/URI; yes	Evidence repository/provider	Resolves to final immutable export; local path stays within approved evidence root	Missing/mutable-only path: archive final export and record path/hash	System generated/exported
signatureAuditTrailPathOrUri	Path/URI; yes	Provider/evidence repository	Resolves to identity, auth, intent and timestamps	No audit trail: approval receives no credit	System generated/exported
signatureEvidenceSha256	Lowercase 64-hex; yes	Exact final evidence bytes	Regex <code>^[a-f0-9]{64}\$</code> and independent recomputation match	Uppercase, malformed or mismatch: recompute exact final bytes; investigate drift	Automatic plus independent verification
signatureCertificateReference	Durable certificate ref; conditional	Digital-signature provider/PKI	Required for certificate-backed signatures; chain and validity check	Missing/invalid chain: reject that signature method	System plus verifier
trustedTimestampReference	Durable timestamp ref; conditional	Trusted timestamp/provider	Required when policy/signature method uses trusted timestamping	Missing/invalid: obtain valid timestamp evidence or use approved method	System plus verifier
evidenceExportedAt	ISO-8601 timestamp; yes	Evidence export service	At/after approval; corresponds to final export	Export predates approval or changed later: export again and rehash	System generated
evidenceRetentionClass	Controlled enum; yes	Records policy; records owner	Approved retention, access, redaction and deletion rules	Undefined class: governance must approve before collection	Governance confirmation

Independent-verification fields

verifiedBy	Stable subject/reference; yes	Verification workflow	Qualified person, distinct where policy requires; no anonymous agent	Missing/same prohibited producer: appoint independent verifier	Independent verification
verifierRole	Exact controlled role; yes	Verification policy/authority roster	Holds verification authority for scope	Ordinary app role: reject; prove verifier appointment	Independent verification
verificationOrganizationId	Stable tenant/org ID; yes	Verifier authority/session	Matches governed organization or approved independent party	Wrong/unapproved org: reject	Independent verification
verifiedAt	ISO-8601 timestamp; yes	Trusted verifier workflow clock	After final evidence export and hash computation	Earlier/missing timestamp: rerun verification	Independent verification
verificationMethod	Controlled enum/string; yes	Verification policy	Documents identity, authority, hash, signature and timing checks	Vague manual review: reject; record reproducible method	Independent verification
verificationResult	Enum PASS or FAIL; yes	Verification engine/person	Must be PASS before import	Fail/unknown: do not import; remediate and reverify	Independent verification
verificationEvidenceReference	Durable ref; yes	Verification record	Resolves to immutable checklist/result	Missing/unresolvable: rerun and export verification evidence	Independent verification
recomputedSignatureEvidenceSha256	Lowercase 64-hex; yes	Independent read of final export	Exact match to signatureEvidenceSha256	Mismatch: quarantine evidence and investigate	Independent verification
contractHashMatch	Boolean; yes	Independent contract hash	Must be true	False: invalidate approval; never rewrite stored hash to match	
roleAuthorityMatch	Boolean; yes	Authority and obligation comparison	Must be true	False: appoint correct role and collect a new approval	
freshAuthenticationWindowValid	Boolean; yes	Auth/approval timestamp comparison	Must be true; zero-to-ten-minute interval	False: new fresh auth and approval required	
signatureReferenceResolved	Boolean; yes	Provider/evidence resolution	Must be true	False: restore durable evidence or recollect approval	
segregationOfDutiesValid	Boolean; yes	Approved SoD policy evaluation	Must be true	False: reassign signer/verifier or obtain permitted governed exception	

Operational fields

fieldOwner	Non-empty function/person reference; yes	Completion plan; governance owner	Names the function responsible for closing the row	Generic team: assign accountable function	Governance confirmation
resolutionAction	Non-empty actionable text; yes	Blocker register; control owner	States exact next action and evidence expected	Vague action: rewrite with owner/output/validation	Automatic draft plus human confirmation
blockerStatus	Controlled status; yes	Completion workflow	Open until all checks pass; closed only after independent verification/import	Premature close: reopen and investigate	Automatic/system
blockingReason	Non-empty text while open; yes	Validation result	Matches failed fields; no hidden failures	Missing/stale reason: regenerate from field results	Automatic/system
invalidationConditions	Non-empty enum array; yes	Policy; governance/control owners	Covers contract drift, authority change, auth/timing, evidence/hash and SoD/review failures	Missing conditions: use approved common list before signing	Governance confirmation
lastValidatedAt	ISO-8601 timestamp; yes	Validation runner clock	Updated only when checks actually run	Invented/stale timestamp: rerun validation	Automatic/system

Current live-validator rules

`scripts/pos-g1-contract-gate.js` currently requires:

- 1 A matching decision record and exact selected option.
- 2 Non-empty rationale and effective version.
- 3 Parseable review/expiry timestamp.
- 4 Non-empty evidence links and affected capabilities.
- 5 Non-empty rollback/disable policy.
- 6 `approvalStatus = APPROVED`.
- 7 At least one approval and complete coverage of every required role.
- 8 Non-empty accountable name, exact role and authority reference.
- 9 Parseable fresh-authentication and approval timestamps.
- 10 Approval at/after fresh authentication and within ten minutes.
- 11 Non-empty signature reference.
- 12 Lowercase 64-character evidence SHA-256.
- 13 Exact register binding to contract artifact, version, path and recomputed hash.

Important checks not enforced by the current script

The script validates shape and timing but does not itself:

- 1 resolve `authorityReference` against an authoritative source;
- 1 prove that `accountableApprover` maps to a stable subject ID;

- 1 verify tenant/organization binding;
- 1 verify authority start/end, revocation, delegation or scope;
- 1 verify reviewer qualification;
- 1 enforce approved same-person/SoD combinations;
- 1 prove MFA or authentication method;
- 1 resolve signatureReference;
- 1 read and rehash the signature evidence bytes;
- 1 validate certificates or trusted timestamps;
- 1 verify post-signature immutability;
- 1 enforce verifier independence;
- 1 detect later authority, policy or evidence revocation.

These checks are mandatory independent pre-import controls even though the narrow script does not inspect them yet.

Automatic versus human responsibility

Read contract IDs/options/roles and compute hash	Yes	Independent review of final binding
Locate candidate names	Yes	Governance decides identity and authority
Assign a G1 authority	No autonomous assignment	Yes, authorized governance owner
Generate fresh-auth timestamp	Trusted auth service only	Signer must actually reauthenticate
Approve or reject	No agent approval	Yes, authorized signer
Export and hash evidence	Yes	Custodian confirms final export
Verify hashes and rules	Yes	Independent verifier accepts responsibility
Set live decision to APPROVED	Controlled import after pass	Human governance approval must already exist

Acceptance rule

A role obligation is acceptable only when every mandatory field passes. A G1 decision is acceptable only when all three exact role obligations pass. G1 is acceptable only when all 11 decisions pass against the unchanged frozen contract hash and the actual gate reports PASSED.